Table 8.6 Economic and sustainability outcomes of UAV-integrated agriculture.

Outcome area	UAV-enabled improvement	Economic benefit	Environmental/ sustainability benefit
Input resource use (fertilizers, pesticides)	Targeted spraying based on vegetation health data	Reduced chemical expenditure and operational cost	Lower soil contamination and prevention of water pollution
Water management	Real-time moisture and thermal mapping for selective irrigation	Reduced water consumption and energy use in pumping systems	Conservation of freshwater resources and improved drought resilience
Crop yield and quality	Early detection of stress, disease, and nutrient deficiencies	Higher and more consistent yields; improved market value	Healthier plant growth with reduced agrochemical dependency
Labor efficiency	Automated field scouting and monitoring flights	Lower labor requirements and reduced human field risk	Reduced manual workload and improved farm safety
Supply chain planning	Accurate yield forecasting and growth progress mapping	Optimized harvest scheduling and reduced post-harvest loss	More stable crop supply with lower waste footprint
Long-term farm sustainability	Continuous multi-season data collection for trend analysis	Data-backed investment and production planning	Builds resilient and climate-adaptive farming systems